commodore semiconductor group

NMOS

MPS
6523
TRI-PORT
INTERFACE

6523 TRI-PORT INTERFACE

CONCEPT . . .

The 6523 TRI-PORT Interface (TPI) is designed to simplify the implementation of complex I/O operations in microcomputer systems. It has three dedicated 8-bit I/O ports which provide 24 individually programmable I/O lines.

FEATURES:

- 24 individually programmable I/O lines
- Completely static operation
- Two TTL Drive Capability
- 6 directly addressable registers
- 1 MHz, 2MHz and 3MHz operation

6523 REGISTERS

*000	R0	PRA — Port Register A
001	R1	PRB — Port Register B
010	R2	PRC — Port Register C
011	R3	DDRA — Data Direction Register A
100	R4	DDRB — Data Direction Register B
101	R5	DDRC — Data Direction Register
110		Illegal States
111		Illegal States

*NOTE: RS2, RS1, RS0 respectively

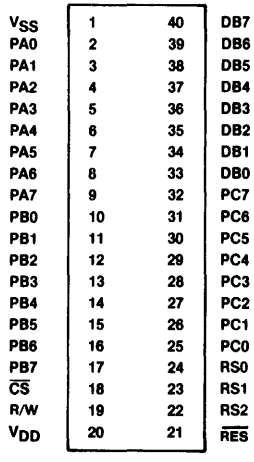
ORDER NUMBER:

MXS 6523

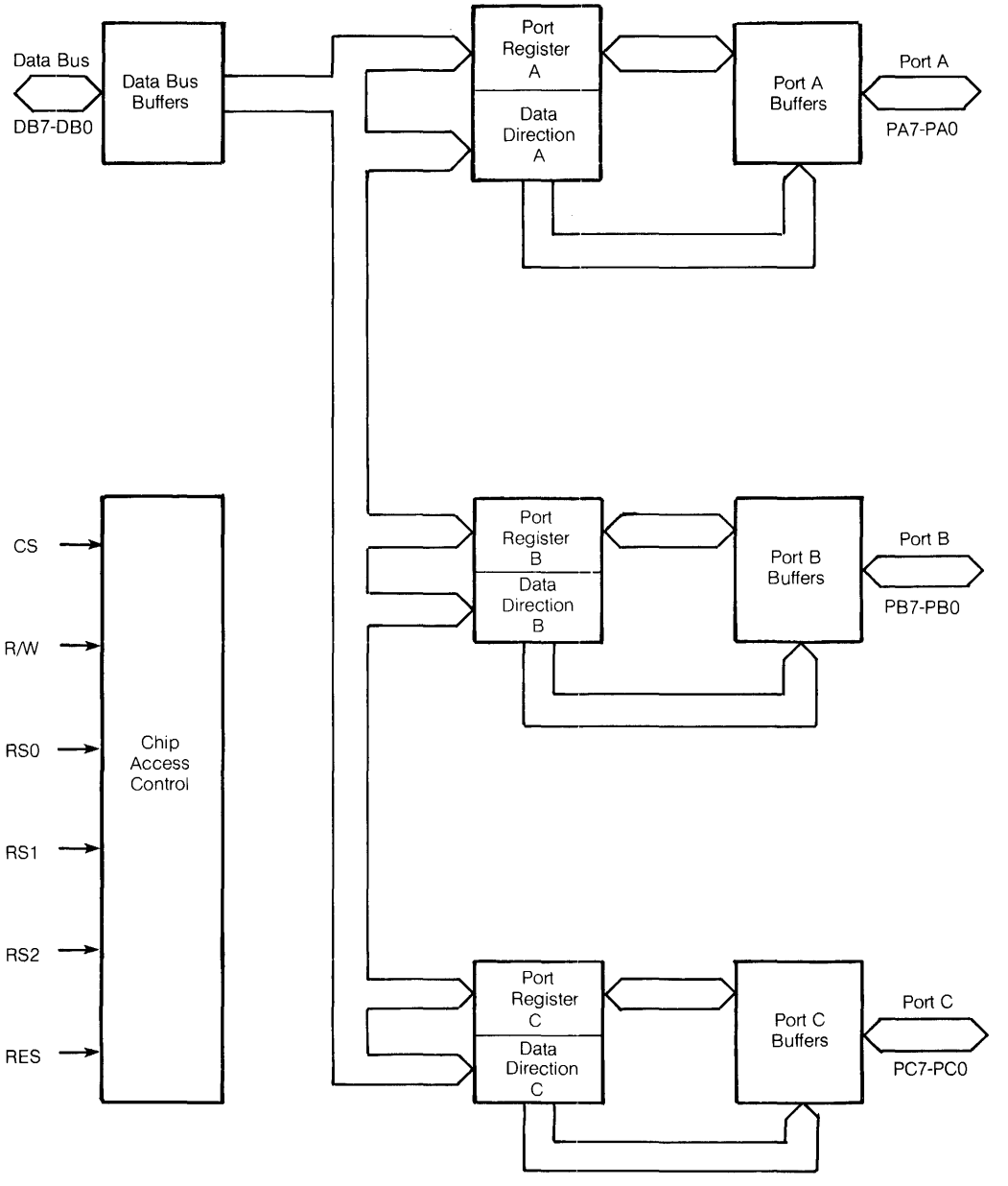
SPEED RANGE
NO SUFFIX = 450 ns
A = 225 ns
B = 165 ns

PACKAGE DESIGNATOR
C = CERAMIC
P = PLASTIC

6523 PIN CONFIGURATION



6523 INTERNAL ARCHITECTURE



MAXIMUM RATINGS

Supply Voltage, V_{CC}	−0.3V to +7.0V
Input/Output Voltage, V_{IN}	−0.3V to +7.0V
Operating Temperature, T_{OP}	0°C to 70°C
Storage Temperature, T_{STG}	−55°C to 150°C

All inputs contain protection circuitry to prevent damage due to high static discharges. Care should be exercised to prevent unnecessary application of voltages in excess of the allowable limits.

COMMENT

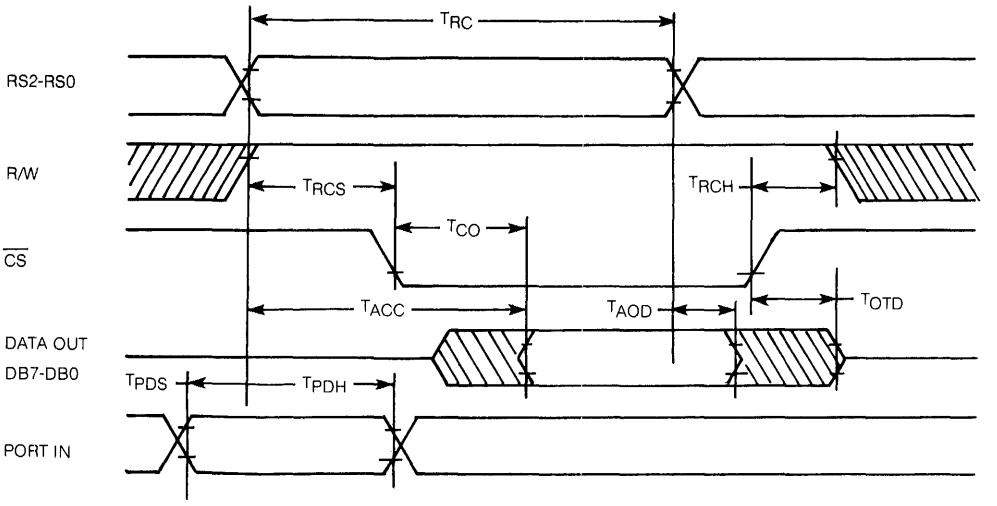
Stresses above those listed under "Absolute Maximum Ratings" may cause permanent damage to the device. These are stress ratings only. Functional operation of this device at these or any other conditions above those indicated in the operational sections of this specification is not implied and exposure to absolute maximum rating conditions for extended periods may affect device reliability.

CHARACTERISTICS (V_{CC} = 5.0 V ± 5%, V_{SS} = 0V, T_A = 0° to 70°C)

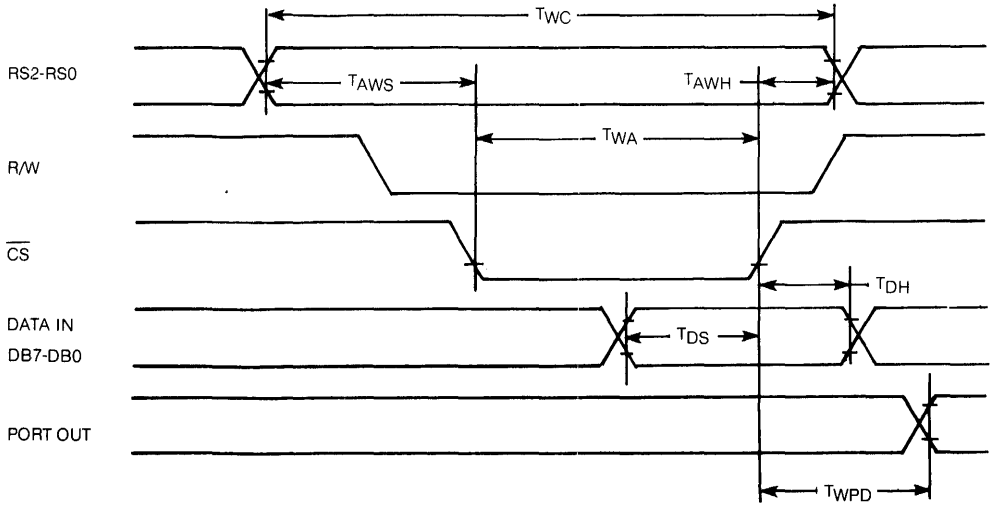
CHARACTERISTIC	SYMBOL	MIN	TYP	MAX	UNIT
Input High Voltage (Normal Operating Levels)	V_{IH}	+ 2.0		V_{CC}	V
Input Low Voltage (Normal Operating Levels)	V_{IL}	−0.3		+0.8	V
Input Leakage Current (V_{in} = 0 to 5.0 V) WRITE, RES, CS, RS2-RS0	I_{IN}	0	± 1.0	± 2.5	μA
Three-State (Off State) Input Current (V_{in} = 0.4 to 2.4 V V_{CC} = max) D0-D7, PA0-P7, PB0-PB7, PC0-PC7	I_{TSI}	0	± 2.0	± 10	μA
Output High Voltage (V_{CC} = min, Load = 200 μA)	V_{OH}	2.4	3.5	V_{CC}	V
Output Low Voltage (V_{CC} = min, Load = 3.2 mA)	V_{OL}	V_{SS}	0.2	0.4	V
Output High Current (Sourcing) (V_{OH} = 2.4 V)	I_{OH}	−200	−1000	—	μA
Output Low Current (Sinking) (V_{OL} = 0.4 V)	I_{OL}	3.2	—	—	mA
Supply Current	I_{CC}	—	50	100	mA
Input Capacitance (V_{in} 0V, T_A = 25°C, f = 1.0 MHz) D0-D7, PA0-PA7, PB0-PB7, PC0-PC7 WRITE, RES, RS2-RS0, CS	C_{in}	—	7	10	pF
Output Capacitance (V_{in} = 0V, T_A = 25°C, f = 1.0 MHz)	C_{out}	—	7	10	pF

Note: Negative sign indicates outward current flow, positive indicates inward flow.

READ CYCLE



WRITE CYCLE



Note: All timings referenced to V_{IL}max, V_{IH} min on inputs and V_{OL} max, V_{OH} min on outputs.

READ CYCLE TIMING

SYMBOL	CHARACTERISTIC	6523		6523A		6523B		UNITS
		MIN	MAX	MIN	MAX	MIN	MAX	
T_{RC}	Read Cycle	450	—	225	—	165	—	ns
T_{ACC}	Access Time[1]	—	450	—	225	—	165	ns
T_{CO}	$\overline{CS}$ to Output Valid	—	270	—	120	—	70	ns
T_{RCS}	R/W high to $\overline{CS}$ Setup	0	—	0	—	0	—	ns
T_{RCH}	R/W high to $\overline{CS}$ Hold	0	—	0	—	0	—	ns
T_{OTD}	$\overline{CS}$ to Output Off Delay	20	120	20	120	20	120	ns
T_{AOD}	Address to Output Delay	50	—	50	—	50	—	ns
T_{PDS}	Port Input Setup	120	—	60	—	40	—	ns
T_{PDH}	Port Input Hold	150	—	150	—	150	—	ns

Note 1: Access Time measured from later of $\overline{WRITE}$ high or RS stable.

WRITE CYCLE TIMING

SYMBOL	CHARACTERISTIC	6523		6523A		6523B		UNITS
		MIN	MAX	MIN	MAX	MIN	MAX	
T_{WC}	Write Cycle	450	—	225	—	165	—	ns
T_{WA}	Write Active Time[2]	420	—	200	—	150	—	ns
T_{AWS}	Address to R/W low Setup	0	—	0	—	0	—	ns
T_{AWH}	Address to R/W low Hold	0	—	0	—	0	—	ns
T_{DS}	Data bus in Setup	150	—	100	—	50	—	ns
T_{DH}	Data bus in Hold	0	—	0	—	0	—	ns
T_{WPD}	Write active to Port out Delay	—	1000	—	500	—	330	ns

Note 2: T_{WA} is the time while both $\overline{CS}$ and R/W are low.

6523 FUNCTIONAL DESCRIPTION

Three 8 bit bi-directional ports (A, B, C) are available on the 6523. Each port has two associated read/write registers:

Data Direction Registers (DDRA, DDRB, DDRC)

Each bit of the data direction registers controls the corresponding pin of the associated port as follows:

DDR bit	Direction of port pin
0	Input (Output driver disabled)
1	Output (Output driver enabled)

Port Registers (PRA, PRB, PRC)

Reading the Port Register returns the logic states of the associated port pins. The pin voltage levels must meet the V_{IH} and V_{IL} specification limits to ensure valid data. (Excessive loading of the output driver may cause the data read to differ from the expected output.) If the port pin is programmed as an output by the DDR, the output driver is set to the last data written to the corresponding PR bit.

6523 INTERFACE SIGNALS

$\overline{CS}$ — Chip Select Input

The $\overline{CS}$ input controls the activity of the 6523. A low level on $\overline{CS}$ causes the device to respond to signals on the R/W and address (RS) lines. A high on $\overline{CS}$ prevents these lines from controlling the 6523. The $\overline{CS}$ line is normally activated (low) by the appropriate address combination from the processor.

R/W — Read/Write Input

The R/W signal is normally supplied by the microprocessor and controls the direction of data transfers of the 6523. A high on R/W indicates a read (data transfer out of the 6523), while a low indicates a write (data transfer into the 6523).

RS_2-RS_0 — Address Inputs

The address inputs select the internal registers (in conjunction with $\overline{CS}$ and R/W) as indicated by the register table.

DB7-DB0 — Data Bus Inputs/Outputs

The eight data bus pins transfer information between the 6523 and the system data bus. These pins are high impedance inputs unless $\overline{CS}$ is low and R/W is high, to read the device. During this read, the data bus output buffers are enabled, driving the data from the selected register onto the system data bus.

$\overline{RES}$ — Reset Input

A low on the RES pin clears internal registers. This sets all three ports as inputs (floating), preventing any conflicts on the bi-directional port lines. For port pins to be used as outputs, the desired output data may be written to the port register before enabling the output driver. This sequence can eliminate undesired output conditions when the outputs are enabled via the DDR.

COMMODORE SEMICONDUCTOR GROUP reserves the right to make changes to any products herein to improve reliability, function or design. COMMODORE SEMICONDUCTOR GROUP does not assume any liability arising out of the application or use of any product or circuit described herein; neither does it convey any license under its patent rights nor the rights of others.